Inductive Link Prediction (ILP) - project assignment

Environment for enhanced inductive link prediction for knowledge graph evaluation using selected NN architectures, which should include:

- The use or adaptation of at least 3 different architectures for a neural network-based prediction model (e.g., based on implementations provided in the PyKEEN library (see below), other libraries, or code accompanying scientific papers),

- A comparison performed on at least 3 datasets (the adaptation of your own dataset prepared for the inductive learning scenario is welcome),

- An analysis of the evaluation methodology used in the state of the art (e.g., in PyKEEN or in https://arxiv.org/abs/1911.06962 / https://github.com/kkteru/grail), including a comparison of different variants for sampling negative triples/examples (e.g., different numbers of examples or different sampling strategies).

Useful links:

- PyKEEN package: A Python package for Knowledge Graphs embeddings:
  - Inductive link prediction tutorial:
    https://pykeen.readthedocs.io/en/stable/tutorial/inductive_lp.html
  - Benchmark inductive datasets internally available
    https://github.com/pykeen/pykeen#inductive-datasets
  - Github:
    https://github.com/pykeen/pykeen
- KG Inductive Link Prediction Challenge (ILPC) 2022:
  - The challenge repository
    https://github.com/pykeen/ilpc2022
  - Paper
    https://arxiv.org/abs/2203.01520
- Additional papers/repositories
  - The methodology and datasets introduction (GraIL)
    https://github.com/kkteru/grail
    https://arxiv.org/abs/1911.06962
  - ILP with hypergraphs (HYPER)
    https://github.com/HxyScotthuang/HYPER
    https://arxiv.org/abs/2506.12362
  - Many other paper
- Course of Jure Leskovec (Stanford)
  - https://snap.stanford.edu/class/cs224w-2025/
    - Inductive vs transductive learning (Lecture 5 from slide 59)
    - Learning with Knowledge Graphs (Lecture 10)